GEOSPATIAL ANALYSIS OF AIR POLLUTION

MINI PROJECT REPORT

Submitted in partial fulfilment of the requirement for the award of the Degree of

MASTER OF SCIENCE IN DATA ANALYTICS

Submitted by

JENSIYA J

24MDA002

Under the Guidance of

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Since 1988

DEPARTMENT OF COMPUTER SCIENCE

CMS COLLEGE OF SCIENCE & COMMERCE

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Coimbatore -640149

NOVEMBER -2025

CERTIFICATE

This is to certify that the Mini Project work entitled

PURE AIR INSIGHTS: ANALYZING POLLUTION PATTERNS

JENSIYA J

24MDA002

in partial fulfillment of the requirement for the award of the degree of

MASTER OF DATA ANALYTICS

Submitted for Viva - Voce Examination held on

Countersigned

Project Guide **Head of the Department**

Internal Examiner **External Examiner**

DECLARATION

I " JENSIYA J [24MDA002]" hereby declare that the project work entitled **"GEOSPATIAL ANALYSIS OF AIR POLLUTION"** developed at CMS College of Science & Commerce and submitted to Bharathiar University, Coimbatore in partial fulfillment of the requirements for the award of degree of Master of Data Analytics is a record of original project work done by me during the period of august-october[2025], under the supervision and guidance of **Dr. SARASWATHI** M.Sc., M.Phil., Ph.D., Associate Professor Department of Computer Science, CMS College of Science & Commerce, Coimbatore.

Place: Coimbatore - 49 Signature of the Student

Date:

ACKNOWLEDGEMENT

First of all I express my immense gratitude to Almighty God for the blessings bestowed on us through my project work. Grateful to **Dr. H. Balakrishnan**, Principal, CMS College for giving me permission to conduct this study.

I wish to express my profound gratitude to **V.Saraswathi**, Head of the Department, Department of Computer Science. I extend my sincere gratitude to guide **Dr.B. Sumathi**, Associate Professor, and Department of Computer Science who encouraged and guided me in every stage of my study.

I extend to her my deepest sense of gratitude for all her sincere effort to guide me in the best possible way. Last but not the least, I am very thankful to all other faculty members of the Department of Computer Science, CMS College of science and commerce. I am also thankful to my friends and all other well wishers who helped me in one way to complete my study.

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1. ABSTRACT

In India, the environmental conditions have been continuously disintegrated because of the industrial developments, urbanization, lack of awareness and motor vehicles. The emissions from industry and vehicles are the prime purpose behind toxins in India. Various pollutants, namely, sulphur dioxide, nitrogen dioxide, carbon monoxide, ground- level ozone (O3), PM10 and PM2.5 impacts the air quality and consequently leads to the effect of human health. Latterly, these specifications were observed to determine the nature of air for the significant number of states and urban communities in India. A report of all stations, urban communities, and states all over India is generated, using AQI (Air Quality Index). The Environmental Protection Agency (EPA) introduced a tool named AQI to assess the levels of pollution in environment because of the major air pollutants.

These findings underscore the urgent need for improved air quality management strategies, including the promotion of public transportation and the adoption of clean energy sources to mitigate adverse health effects and protect public well-being and the study provides critical data for policymakers to develop targeted interventions to protect public health from the harmful impacts of air pollution.

Hence the clean and fresh air is a sole requirement for good human health and wellbeing of the humanity. Several programs have begun to keep track on the quality level of air and to accumulate the vast amount of data on concentration of each air pollutant (e.g., SPM, CO, NO2, SO2, etc.) in different parts of the world. The massive data usually don't pass on the air quality status to the scientific community, officialdom, policy makers, and especially to the overall public during a simple and straightforward manner. This issue is approached by deciding the Air Quality Index (AQI) of guaranteed area using Data Sets and Power Bi tools.

2. INTRODUCTION

This project presents a study of analyzing the Air Quality Index using individual methods and tools and to interpret the significance of AQI in atmosphere. This empowers us to know the perception of various creators and gives a clear view of data interpretation and comparison between the various tools. Analysis of Air Quality Index – In this work the authors have proposed a detailed research on Air Quality Index. They analyzed the Air Quality Index Values for three cities Delhi, Bangalore and Chennai. They found Air quality Index can be calculated by two methods, one as taking the greatest value among the pollutants and the other is finding the average of the considerable number of poisons. Also this paper predicted the AQI for the future years using Regression analysis method. All the pollutants SO2, NO2, RSPM, and PM are analyzed with analysis of Variance test. The classification for the calculated Air Quality Index values are as Good (0-50), Moderate (51-100), Unhealthy for sensitive groups (101-150), Unhealthy (151-200), Very unhealthy (210-300), and Hazardous (301-500).

It is information perception and Business insight that converts data from different data sources to BI reports Power BI suite gives numerous software, connector, and services - Power BI desktop, Power BI service dependent on Dataset, and portable Power BI applications accessible for various platforms These arrangement of services are utilized by business clients to consume information and manufacture BI reports.

From Visualizations, select table chart and select required items for showing the top 10 cities with highest population. Similarly, select another table chart and select required objects for displaying top 10 cities with lowest population. Select Maps from Visualizations and select required objects in order to display all the cities in the map. Apply all the required filters in order to view the map and understand easily. Similarly display the top 10 cities with highest health risks and lowest health risks.

TYPES OF DATA:

STRUCTURED DATA

Structured data is the data which conforms to a data model, has a well define structure, follows a consistent order and can be easily accessed and used by a person or a computer program. Structured data is usually stored in well-defined schemas such as Databases. It is generally tabular with column and rows that clearly define its attributes. SQL (Structured Query language) is often used to manage structured data stored in databases.

SEMI-STRUCTURED DATA

Semi-structured data is data that does not conform to a data model but has some structure. It lacks a fixed or rigid schema. It is the data that does not reside in a rational database but that have some organizational properties that make it easier to analyze with some processes, we can store them in the relational database.

- Data does not conform to a data model but has some structure.
- Data cannot be stored in the form of rows and columns as in Databases.
- Semi-structured data contains tags and elements (Metadata) which is used to group data and describe how the data is stored.
- Similar entities are grouped together and organized in a hierarchy.
- Entities in the same group may or may not have the same attributes or properties.
- Does not contain sufficient metadata which makes automation and management of data difficult.
- Size and type of the same attributes in a group may differ.
- Due to lack of a well-defined structure, it cannot used by computer programs easily.

UNSTRUCTURED DATA

Unstructured data is the data which does not conforms to a data model and has no easily identifiable structure such that it cannot be used by a computer program easily. Unstructured data is not organized in a pre-defined manner or does not have a pre- defined data model, thus it is not a good fit for a mainstream relational database.

- Data neither conforms to a data model nor has any structure.
- Data cannot be stored in the form of rows and columns as in Databases.
- Data does not follow any semantic or rules.
- Data lacks any particular format or sequence.
- Data has no easily identifiable structure.
- Due to lack of identifiable structure, it cannot used by computer programs easily.

MACHINE LEARNING

Machine Learning is the field of study that gives computers the capability to learn without being explicitly programmed. ML is one of the most exciting technologies that one would have ever come across. As it is evident from the name, it gives the computer that makes it more similar to humans: The ability to learn.

Machine learning is an important component of the growing field of data science. Through the use of statistical methods, algorithms are trained to make classifications or predictions, and to uncover key insights in data mining projects. These insights subsequently drive decision making within applications and businesses, ideally impacting key growth metrics. As big data continues to expand and grow, the market demand for data scientists will increase. They will be required to help identify the most relevant business questions and the data to answer them. Machine learning algorithms are typically created using frameworks that accelerate solution development, such as TensorFlow and PyTorch

METHODOLOGY FOR AIR POLLUTION ANALYS

Carbon Dioxide (CO2) - This is by far the most emitted form of human caused air pollution. The CO2 increase in earth's atmosphere has been accelerating.

Sulfur Dioxide (SO2) - SO2 is produced by volcanoes and in various industrial processes. Coal and petroleum often contain sulfur compounds, and their combustion generates sulfur dioxide. Further oxidation of SO2, usually in the presence of a catalyst such as NO2, forms H2SO4, and thus acid rain. This is one of the causes for concern over the environmental impact of the use of these fuels as power sources.

Nitrogen Dioxide (NO2) - Nitrogen dioxide is a chemical compound with the formula NO2. It is expelled from high temperature combustion, and is also produced during thunderstorms by electric discharge. They can be seen as a brown haze dome above or a plume downwind of cities. It is one of the most prominent air pollutants; this reddish-brown toxic gas has a characteristic sharp, biting odor.

Carbon Monoxide (CO) - CO is a colorless, odorless, toxic yet non-irritating gas. It is a product of incomplete combustion of fuel such as natural gas, coal or wood. Vehicular exhaust is a major source of carbon monoxide.

Particulate Matter (PM) - Particulates, alternatively referred to as particulate matter (PM), atmospheric particulate matter, or fine particles, are tiny particles of solid or liquid suspended in a gas. In contrast, aerosol refers to combined particles and gas. Some particulates occur naturally, originating from volcanoe2s, dust storms, forest and grassland fires, living vegetation, and sea spray. Human activities, such as the burning of fossil fuels in vehicles, power plants and various industrial processes also generate significant amounts of aerosols. **Ozone (O3)** – O3 is a key constituent of the troposphere. Ground level ozone (O3) is formed from NOx and VOCs. It is also an important constituent of certain regions of the stratosphere commonly known as the Ozone layer. Photochemical and chemical reactions involving it drive many of the chemical processes that occur in the atmosphere by day and by night.

3. DATASET

Sampling Date	State	City/Town/ Village/Area	Location of Monitoring Station	Agency	Type of Location	SO2	NO2	RSPM/PM10	SPM
01-05-21	Tamil Nadu	Chennai	Kathivakkam, Municipal Kalyana Mandapam, Chennai	Tamilnadu State Pollution Control Board	Industrial Area	9.6	17.16667	73.33333333	149.666667
01-07-2	Tamil Nadu	Chennai	Kathivakkam, Municipal Kalyana Mandapam, Chennai	Tamilnadu State Pollution Control Board	Industrial Area	11.15	20.28333	61.33333333	150.333333
18-08-21	Tamil Nadu	Coimbatore	Distt. Collector's Office, Coimbatore	Tamilnadu State Pollution Control Board	Industrial Area	4	20.88	41.33333333	157.666667
25-08-21	Tamil Nadu	Coimbatore	Distt. Collector's Office, Coimbatore	Tamilnadu State Pollution Control Board	Industrial Area	4	15.55	94.5	106.5
25-09-21	Tamil Nadu	Madurai	Fenner (I) Ltd. Employees Assiciation Building Kochadai, Madurai	Tamilnadu State Pollution Control Board	Industrial Area	10.575	26.15	28	61.5
29-09-21	Tamil Nadu	Madurai	Fenner (I) Ltd. Employees Assiciation Building Kochadai, Madurai	Tamilnadu State Pollution Control Board	Industrial Area	13.05	21.075	27.5	89.5
20-01-21	Tamil Nadu	Salem	Sowdeswari College Building, Salem	Tamilnadu State Pollution Control Board	Industrial Area	7.2	22.76667	112	123

03-08-21	Tamil Nadu	Thoothukudi	Fisheries College, Tuticorin	Tamilnadu State Pollution Control Board	Industrial Area	9.6	17.16667	73.333333333	149.6666667
03-11-21	Tamil Nadu	Thoothukudi	Fisheries College, Tuticorin	Tamilnadu State Pollution Control Board	Industrial Area	11.15	20.28333	61.333333333	150.3333333
29-11-22	Tamil Nadu	Trichy	Central Bus Stand, Trichy	Tamilnadu State Pollution Control Board	Residential, Rural and other Areas	4	20.88	41.333333333	157.6666667
12-04-22	Tamil Nadu	Trichy	Central Bus Stand, Trichy	Tamilnadu State Pollution Control Board	Residential, Rural and other Areas	4	15.55	94.5	106.5
24-12-23	Tamil Nadu	Mettur	SIDCO Industrial Complex, Mettur	Tamilnadu State Pollution Control Board	Industrial Area	10.575	26.15	28	61.5
27-12-23	Tamil Nadu	Mettur	SIDCO Industrial Complex, Mettur	Tamilnadu State Pollution Control Board	Industrial Area	13.05	21.075	27.5	89.5
20-08-23	Tamil Nadu	Cuddalore	SIPCOT Industrial Complex, Cuddalore	Tamilnadu State Pollution Control Board	Industrial Area	7.2	22.76667	112	123
27-05-24	Tamil Nadu	Chennai	Thiruvottiyur Municipal Office, Chennai	Tamilnadu State Pollution Control Board	Industrial Area	9.6	17.16667	73.333333333	149.6666667
30-05-24	Tamil Nadu	Chennai	Thiruvottiyur Municipal Office, Chennai	Tamilnadu State Pollution Control Board	Industrial Area	11.15	20.28333	61.333333333	150.3333333
06-03-24	Tamil	Chennai	Thiruvottiy	Tamilnadu	Industrial	4	20.8	41.33	157.

	Nadu		ur Municipal Office, Chennai	State Pollution	Area		8	33333 3	666 666 7
16-07-24	Tamil Nadu	Salem	Sowdeswa ri College Building, Salem	Tamilnadu State Pollution Control Board	Residenti al, Rural and other Areas	4	15.5 5	94.5	106. 5
18-07-24	Tamil Nadu	Salem	Sowdeswa ri College Building, Salem	Tamilnadu State Pollution Control Board	Residenti al, Rural and other Areas	10.5 75	26.1 5	28	61.5
21-07-24	Tamil Nadu	Salem	Sowdeswa ri College Building, Salem	Tamilnadu State Pollution Control Board	Residenti al, Rural and other Areas	13.0 5	21.0 75	27.5	89.5
10-04-23	Tamil Nadu	Coimbator e	SIDCO Office, Coimbator e	Tamilnadu State Pollution Control Board	Industrial Area	7.2	22.7 6667	112	123
17-04-22	Tamil Nadu	Coimbator e	SIDCO Office, Coimbator e	Tamilnadu State Pollution Control Board	Industrial Area	4	15.5 5	94.5	106. 5
05-05-24	Tamil Nadu	Coimbator e	SIDCO Office, Coimbator e	Tamilnadu State Pollution Control Board	Industrial Area	10.5 75	26.1 5	28	61.5
23-12-23	Tamil Nadu	Trichy	Central Bus Stand, Trichy	Tamilnadu State Pollution Control Board	Residenti al, Rural and other Areas	13.0 5	21.0 75	27.5	89.5
30-12-24	Tamil Nadu	Trichy	Central Bus Stand, Trichy	Tamilnadu State Pollution Control Board	Residenti al, Rural and other Areas	7.2	22.7 6667	112	123

4. NORMALIZATION

DATA NORMALIZATION

Normalization is the process of reorganizing data in a database so that it meets two basic requirements:

1. There is no redundancy of data, all data is stored in only one place.

2. Data dependencies are logical ,all related data items are stored together.

Data normalization is an essential step in the geospatial analysis of air pollution to ensure consistency and comparability among datasets collected from various sources such as satellite sensors, monitoring stations, and meteorological data. Since these datasets often differ in units, scales, and ranges, normalization converts them into a common scale, reducing bias and improving analytical accuracy. Techniques such as min–max scaling or z-score normalization are used to standardize pollutant concentrations like PM2.5, NO_2 , and SO_2 . This process enables effective spatial modeling, mapping, and integration of multiple data layers, leading to more reliable visualization and interpretation of air pollution patterns across regions.

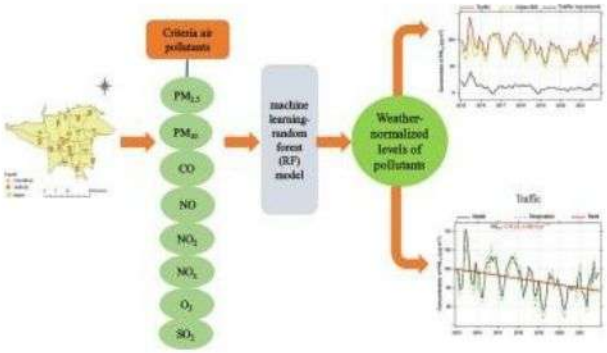
DATA NORMALIZATION TECHNIQUES:

The most widely used types of normalization in machine learning are:

Min-Max Scaling – Subtract the minimum value from each column's highest value and divide by the range. Each new column has a minimum value of 0.

Standardization Scaling – The term standardization refers to the process of centering a variable at zero and standardizing the variance at one. Subtracting the mean of each observation and then dividing by the standard deviation is the procedure:

The features will be rescaled so that they have the attributes of a typical normal distribution with standard deviations.



5. ANALYSIS AND MODELLING

DATA PREPROCESSING

In the data preprocessing in Power BI plays a crucial role in ensuring accurate visualization and analysis. Raw datasets containing air quality metrics, geographic coordinates, and temporal data are cleaned, transformed, and integrated using Power Query. Missing values are handled, data types are standardized, and irrelevant fields are removed to enhance performance. Spatial data is then mapped using Power BI's geospatial features, enabling interactive visualizations that reveal pollution patterns across regions and time periods. This preprocessing stage ensures reliable and insightful analysis for effective decision-making.

DATA PREPROCESSING STEPS

The steps used in data preprocessing include the following:

1. **Data cleansing**. The aim here is to find the easiest way to rectify quality issues, such as eliminating bad data, filling in missing data or otherwise ensuring the raw data is suitable for feature engineering.
2. **Data reduction**. Raw data sets often include redundant data that arise from characterizing phenomena in different ways or data that is not relevant to a particular ML, AI or analytics task. Data reduction uses techniques like principal component analysis to transform the raw data into a simpler form suitable for particular use cases.
3. **Data transformation**. Here, data scientists think about how different aspects of the data need to be organized to make the most sense for the goal. This could include things like structuring unstructured data, combining salient variables when it makes sense or identifying important ranges to focus on.
4. **Data validation**. At this stage, the data is split into two sets. The first set is used to train a machine learning or deep learning model. The second set is the testing data that is used to gauge

the accuracy and robustness of the resulting model.

The above data containing air pollution data set is already preprocessed. It contains 1172 values. Many missing values are present after preprocessing it becomes 652.It deletes the unwanted columns. All the dataset are preprocessed. Hence there is no need for further preprocessing the data.

ANALYSING

In this project, a simple machine learning approach was used to predict air pollution levels in Tamil Nadu for the next five years. Historical air quality data, including parameters such as PM2.5, PM10, NO_2 , and SO_2 , were analyzed to understand past trends and patterns. A regression-based algorithm, such as Linear Regression, was applied to forecast future pollution levels. The model's predictions were visualized in Power BI to display expected changes across regions, helping identify areas that may face higher pollution risks in the future and supporting environmental planning and policy decisions.

MODELING

In this project, single-variable linear regression was applied to analyze and predict air pollution trends based on historical data. The method uses one independent variable, such as time or a specific pollutant concentration, to model its relationship with overall air quality levels. By fitting a best-fit line through the data, the regression model helps identify patterns, trends, and possible future values of air pollution. This simple yet effective approach provides insights into how pollution levels change over time and serves as a foundational step for more advanced predictive modeling and environmental decision-making.

Data Preparation: Ensure you have a dataset with relevant variables, including the independent variable and the dependent variable. Clean and preprocess the data as needed, handling missing values, outliers, and formatting issues.

Data Visualization: Create scatter plots to visualize the relationship between the independent and dependent variables. Use Power BI's visualization tools to add trendlines and calculate correlation coefficients to assess the accuracy of the pollution.

Dashboard Integration: Integrate the results of the linear regression analysis into your Power BI dashboard. Create visualizations to display the regression equation, coefficients, and diagnostic plots. Provide interactive features to allow users to explore the relationship between variables and visualize the impact of different factors on coffee shop performance.

6. VISUALIZATION OF DATA

This project focuses on the geospatial analysis and visualization of air pollution data to understand spatial patterns and identify pollution hotspots across different regions. Air pollution is a critical environmental issue affecting public health and climate conditions. By visualizing air quality data on a map, meaningful insights can be derived to support environmental management and policy decisions.

In air pollution simulation using Power BI, visualization plays a crucial role in analyzing data, understanding trends, and communicating insights effectively. The following visualization techniques can be applied to monitor, explore, and interpret air quality data:

1. Heat Maps

- Represent pollutant concentrations across different geographic regions.
- Highlight areas with high levels of PM2.5, PM10, NO_2 , SO_2 , or CO
- Use color gradients to quickly identify pollution hotspots.

2. Bubble or Scatter Maps

- Plot monitoring stations or sensors on a map.
- Bubble size or color intensity can indicate pollutant levels at each location.
- Helps detect patterns in spatial pollution distribution.

3. Time-Series Charts

- Track pollutant levels over time to observe daily, weekly, or seasonal trends.
- Useful for detecting peak pollution periods and understanding temporal changes.

4. Gauge or KPI Visuals

- Display current pollutant levels against standard thresholds or safe limits.
- Provides a quick overview of air quality status for specific regions.

5. Cluster Analysis Visualization

- Group areas with similar pollution patterns using clustering techniques like K-Means.
- Identifies zones with similar environmental risks and aids in targeted interventions.

6. Comparative Bar/Column Charts

- Compare pollutant levels across different regions, time periods, or pollutant types.
- Enables easy assessment of pollution intensity and trend comparisons.

7. Interactive Dashboards

- Combine maps, charts, and KPIs in a single interactive dashboard.
- Include slicers or filters to explore data by location, pollutant type, or time.
- Facilitates scenario analysis and decision-making for policymakers or environmental planners.

By implementing these visualizations in Power BI, air pollution data can be interpreted efficiently, enabling effective monitoring, reporting, and predictive analysis to support environmental management strategies.

7. IMPLEMENTATION

The implementation of the project involves integrating data collection, preprocessing, analysis, and visualization within a unified workflow. Air quality data is gathered from various sources such as government monitoring stations, IoT sensors, or public datasets containing pollutant concentrations (e.g., PM2.5, PM10, NO_2 , SO_2 , CO, and O_3). The collected data undergoes preprocessing in Power BI or other analytical tools, including data cleaning, handling missing values, and normalization techniques like **Min-Max scaling** to ensure consistency. Geospatial analysis is performed by mapping pollution levels across different regions to identify hotspots and spatial trends. Visualization dashboards are then developed in **Power BI**, utilizing charts, heat maps, and interactive filters to present insights clearly. This implementation enables effective monitoring, comparative regional analysis, and data-driven decision-making for improving air quality management.

1. Data Collection: Air pollution data is collected from government monitoring stations, satellite imagery, and IoT-based air sensors. The dataset includes pollutants such as PM2.5, PM10, NO_2 , SO_2 , CO, and O_3 , along with geographic coordinates and timestamps.

2. Data Preprocessing: The collected data is cleaned to remove inconsistencies, handle missing values, and standardize units Normalization techniques like Min-Max scaling are applied to ensure uniformity across different pollutant parameters.

3. Geospatial Integration: The preprocessed data is integrated with spatial layers such as city boundaries, land use, traffic density, and population distribution using tools like Power BI or QGIS.

4. Spatial Analysis: Geospatial methods such as heat mapping, spatial clustering, and interpolation (IDW or Kriging) are used to identify pollution hotspots and estimate values in unsampled areas.

5. Visualization: Interactive dashboards and geospatial maps are created in **Power BI** to visualize spatial and temporal pollution trends. These visualizations highlight areas of concern and enable comparative regional analysis.

6. Decision Support: The analysis results assist policymakers and researchers in identifying high-risk zones, tracking pollution sources, and developing effective air quality management strategies.

Building an Air Pollution Dashboard in Power BI

1. Data Import and Connection: Air quality data from various sources such as CSV files, APIs, or databases is imported into Power BI. The dataset includes pollutant levels (PM2.5, PM10, NO_2 , SO_2 , CO, and O_3), along with location coordinates, date, and time.

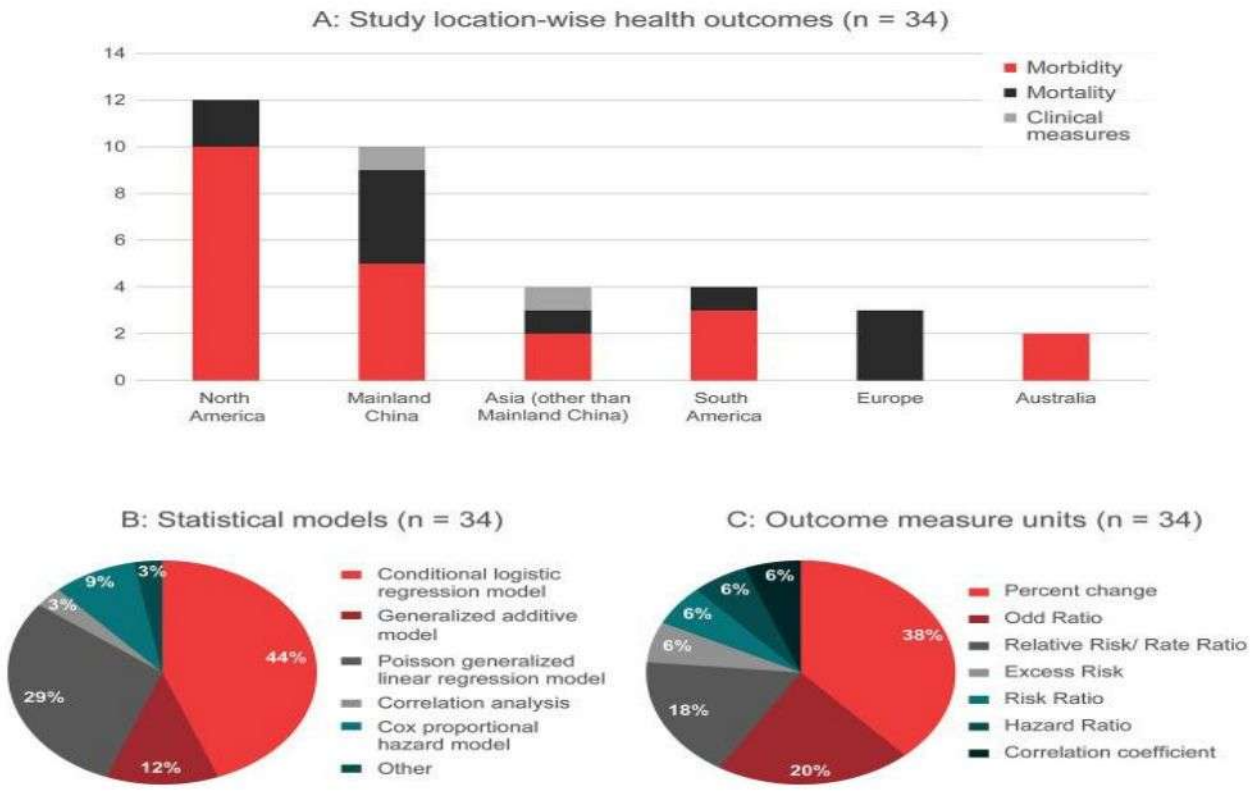
2. Data Cleaning and Transformation: Using Power Query, the data is cleaned to remove missing or duplicate values and standardized using transformations. Columns are formatted for consistency, and data normalization** (e.g., Min-Max scaling) ensures comparable pollutant ranges.

3. Geospatial Mapping: Power BI's Map and Filled Map visuals are used to represent pollutant concentration across different regions. Geographic fields such as latitude, longitude, or city names are used to create interactive geospatial visualizations and heat maps showing pollution intensity.

4. Trend and Comparison Analysis: Line charts, bar graphs, and time-series visuals are added to display daily, monthly, and seasonal variations in pollution levels. Users can compare pollutant concentrations across locations and time periods.

5. Dashboard Interactivity: Slicers, filters, and dropdowns are implemented to enable users to explore specific pollutants, regions, or timeframes dynamically. Tooltips are added to provide quick insights on pollutant levels for each area.

6. Insight Generation: The dashboard summarizes key findings such as highest and lowest pollutant levels, pollution hotspots, and trends over time. These insights help policymakers, researchers, and the public understand air quality patterns and make informed decisions.



8. SCOPE FOR FURTHER DEVELOPMENT

Advanced predictive modeling

A key future direction is the use of machine learning (ML) and deep learning (DL) to forecast air pollution levels with high spatial and temporal precision.

- **Project idea**: Develop an ML-driven framework to predict daily or hourly PM2.5 and ozone levels in urban areas. The framework would use various data inputs, such as satellite imagery (e.g.,

Sentinel-5P), meteorological data (wind, temperature, humidity), traffic patterns, land use information, and historical air quality data.

- **Methodology**: Utilize hybrid models, such as combining convolutional neural networks (CNNs) for spatial pattern recognition with long short-term memory (LSTM) networks for time-series forecasting.

- **Future scope**: Scale the model to different cities or regions, refine it to incorporate real-time sensor data from IoT networks, and use it to simulate the impact of different policy interventions, such as low-emission zones.

Health risk assessment and epidemiology

Future projects can move beyond simply mapping pollutants to correlating them with public health outcomes.

- **Project idea**: Conduct a geospatial analysis linking air pollution exposure to specific health outcomes, like respiratory diseases, cardiovascular events, or neurodegenerative diseases.

- **Methodology**: Create detailed exposure maps by integrating pollutant data with demographic data, vulnerability indices (e.g., age, poverty), and epidemiological data from health records. Use GIS to identify high-risk zones and populations.

- **Future scope**: Integrate real-time air quality data with health databases to trigger targeted public health alerts. Develop tools for "health geomatics" to enable deeper understanding of environmental risk patterns for epidemiological studies.

Integration of IoT and citizen science

The proliferation of low-cost sensors and mobile apps provides an opportunity for more granular, real-time data collection.

- **Project idea**: Create an integrated system that combines data from a network of low-cost, ground-based IoT sensors with satellite and traditional monitoring data.

- **Methodology**: Use geospatial techniques like spatial interpolation (e.g., Inverse Distance Weighted) and kriging to generate high-resolution air quality surface models. A user-friendly mobile app could enable citizen participation by allowing users to contribute data and access localized air quality information.

- **Future scope**: Expand the sensor network to a city-wide scale and integrate the data stream with AI models for real-time monitoring and alerting.

Source apportionment and policy evaluation

Advanced geospatial analysis can be used to pinpoint the sources of pollution and evaluate the effectiveness of mitigation strategies.

- **Project idea**: Develop a model to perform source apportionment, identifying the relative contributions of various sources (e.g., traffic, industry, specific industrial sites) to overall air pollution.
- **Methodology**: Utilize spatiotemporal analysis to correlate pollution hotspots with geospatial data on industrial locations, traffic flows, and land use. Analyze changes in pollutant levels in response to policy changes, such as lockdown periods.
- **Future scope**: Use geospatial models to forecast the impact of future urban development, traffic changes, and industrial zoning on air quality. This can provide a valuable baseline for policymakers to evaluate the success of new regulations.

3D and vertical spatial modeling

While most analysis focuses on 2D maps, future projects can extend into the vertical dimension to better understand pollution dispersion in complex urban environments.

- **Project idea**: Develop a 3D geospatial model of air pollution in street canyons and highly dense urban areas.
- **Methodology**: Integrate digital elevation models (DEMs) and urban building data into GIS to simulate the dispersion of pollutants. Use data from LIDAR and other vertical monitoring systems to inform the model.
- **Future scope**: Extend the 3D models to create real-time simulations that predict how pollution moves and is trapped within different urban geometries, helping city planners design healthier urban spaces.

9. CONCLUSION

WHO has identified in air pollution the second highest risk factor for non-communicable diseases worldwide, with consequent deaths counting to millions. Coherently, the international scientific community have spent extensive efforts in studying the causal association between human health and air pollution, both for short (in days) and long-term (in months to years) exposure. As a

consequence, the existence of a relationship between short-term air pollution exposure and CVD risk is now well-assessed. However, recent technological advancements in data collection and processing are giving a renovated momentum to this field, enabling novel experimental set-ups and new methodological approaches, providing more robust and meaningful results. In this context, we hypothesized an undergoing shifting of the state-of-art, with a possible separation between 'traditional' studies and the ones using advanced geospatial modelling approaches.

Coherently, we performed this systematic review about studies published in the last six years that analysed the effects of short-term exposure of air pollution on acute CVD events, focusing the selection to studies where advanced geospatial techniques were applied in modelling both the exposure and the risk, and where social and environmental determinants of health inequalities that could modify the effect were accounted for. It led to the identification and analysis of 34 studies. The review enlightened a worldwide increasing trend in the use of data sources coming from advanced technologies, as well as in the implementation of state-of-art modelling, with regards to atmospheric pollution. Also, it was possible to highlight that the inclusion of unconventional socio-demographic and environmental confounders, along with personal clinical variables, was found able to modify the effects of the CVD risk, thus helping in better understanding the spatial and structural inequalities of CV health among the studied cities and regions. The discussion on emerging frameworks and relevant confounders addressed the knowledge gap that prevails between public and urban health, and it expects to encourage more transdisciplinary research and comprehensive decision-making in this domain. In particular, the need for a definition of a common standard to be utilized in such studies, for example by identifying a minimal set of confounding factors that should always be included, or by considering a certain way of reporting results to facilitate future meta-analysis, should be addressed by the scientific community.

10. BIBLIOGRAPHY & APPENDICES

BOOK REFERENCE

1. Smith, John. "Building Interactive Dashboards with Power BI".

2. 2. Jones, Sarah. "Data Visualization Best Practices for Power BI Dashboards".

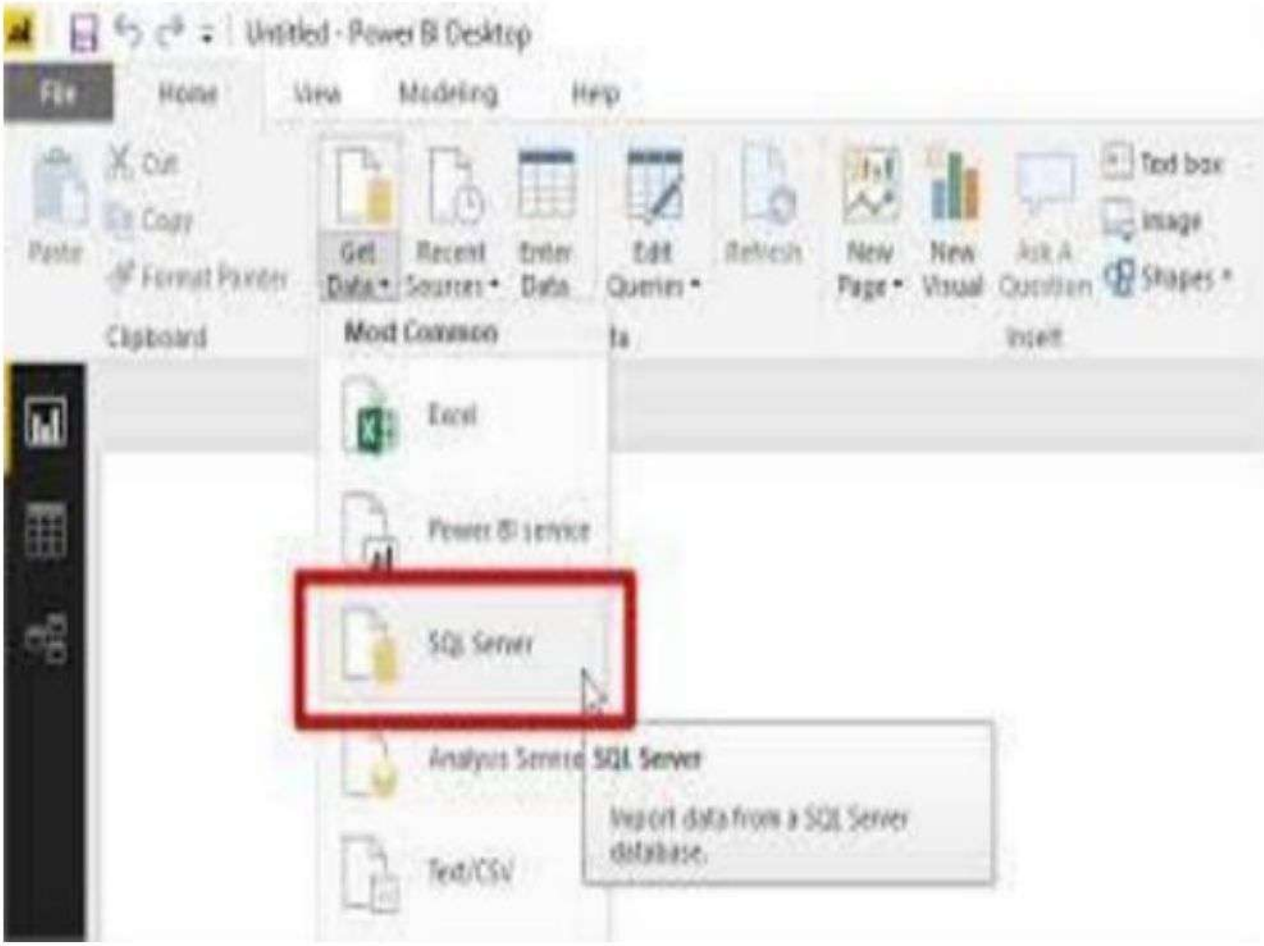
WEBSITE REFERENCE

https://mavenanalytics.io/data-playground?page=3

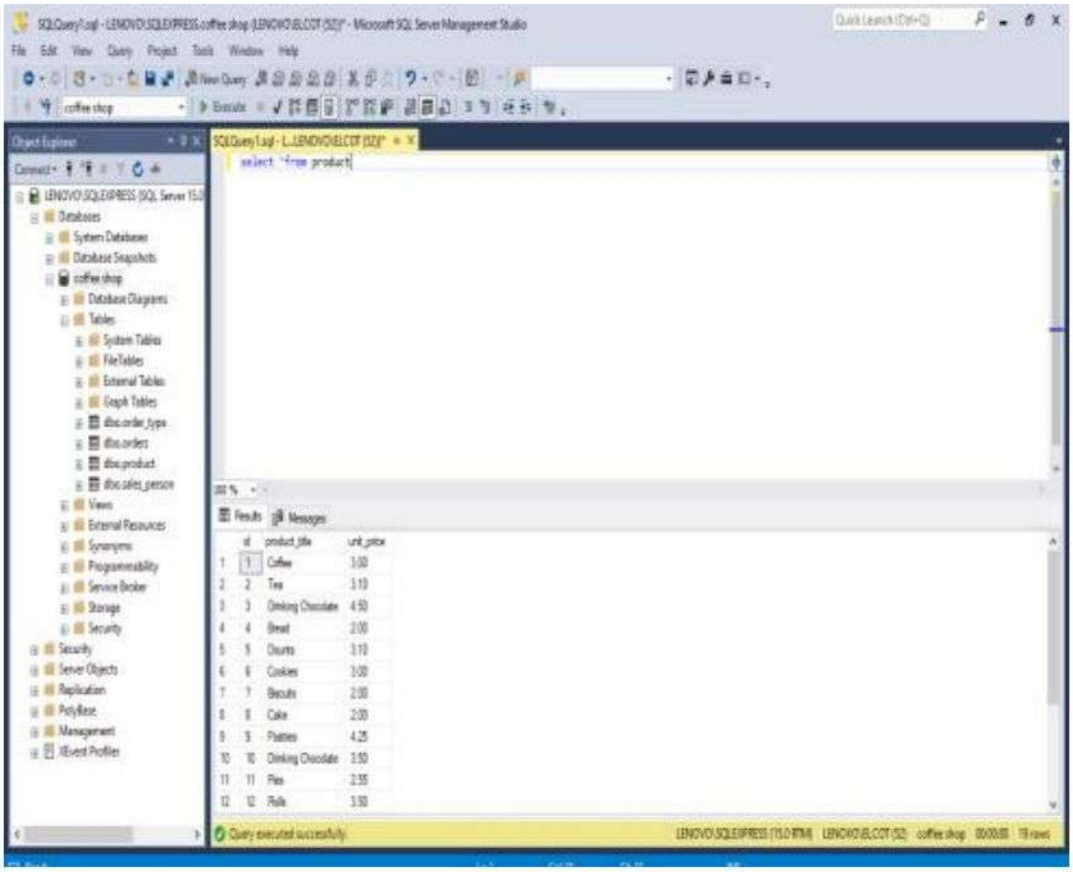
https://www.youtube.com/watch?v=J9K3ZZa5gBM

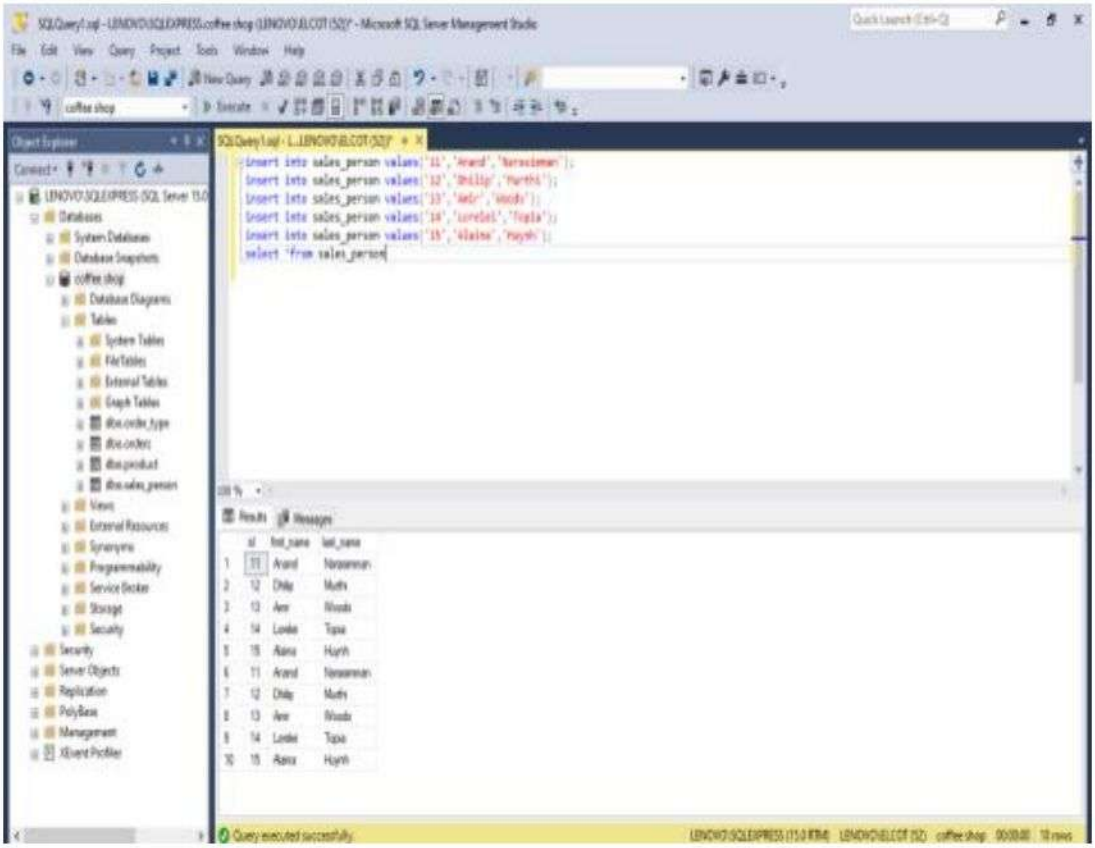
APPENDICES

A.DATA FLOW DIAGRAM



B.SCREEN SHOT


SQLQuery1.sql - LENOVO\SQLEXPRESS.coffee shop (LENOVO\ELCOT (52))* - Microsoft SQL Server Management Studio
File Edit View Query Project Tools Window Help
coffee shop
Execute
Object Explorer
Connect
LENOVO\SQLEXPRESS (SQL Server 15.0
Databases
System Databases
Database Snapshots
coffee shop
Database Diagrams
Tables
System Tables
FileTables
External Tables
Graph Tables
dbo.order_type
dbo.orders
dbo.product
dbo.sales_person
Views
External Resources
Synonyms
Programmability
Service Broker
Storage
Security
Security
Server Objects
Replication
PolyBase
Management
XEvent Profiler
SQLQuery1.sql - L...LENOVO\ELCOT (52))*
select *from product
Results
Messages
id product_title unit_price
1 1 Coffee 3.00
2 2 Tea 3.10
3 3 Drinking Chocolate 4.50
4 4 Bread 2.00
5 5 Donuts 3.10
6 6 Cookies 3.00
7 7 Biscuits 2.00
8 8 Cake 2.00
9 9 Pastries 4.25
10 10 Drinking Chocolate 3.50
11 11 Pies 2.55
12 12 Rolls 3.50
Query executed successfully.
LENOVO\SQLEXPRESS (15.0 RTM) LENOVO\ELCOT (52) coffee shop 00:00:00 19 rows


SQLQuery1.sql - LENOVO\SQLEXPRESS.coffee shop (LENOVO\ELCOT (52))* - Microsoft SQL Server Management Studio
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dbo.sales_person
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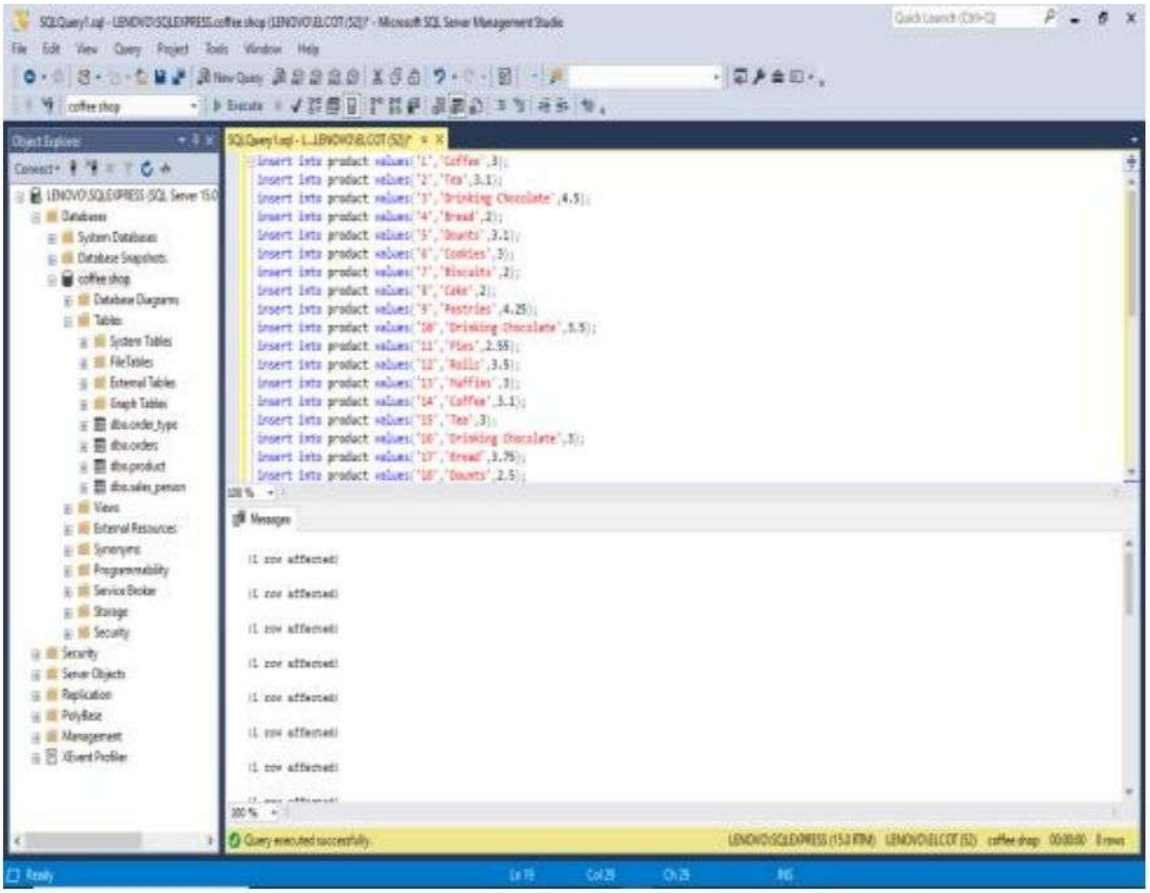
Programmability
Service Broker
Storage
Security
Security
Server Objects
Replication
PolyBase
Management
XEvent Profiler
SQLQuery1.sql - L...LENOVO\ELCOT (52))*
insert into sales_person values('11','Anand','Narasimman');
insert into sales_person values('12','Philip','Murthi');
insert into sales_person values('13','Amir','Moosa');
insert into sales_person values('14','Loveline','Tupia');
insert into sales_person values('15','Alaina','Huynh');
select *from sales_person
Results
Messages
id first_name last_name
1 11 Anand Narasimman
2 12 Philip Murthi
3 13 Amir Moosa
4 14 Loveline Tupia
5 15 Alaina Huynh
6 11 Anand Narasimman
7 12 Philip Murthi
8 13 Amir Moosa
9 14 Loveline Tupia
10 15 Alaina Huynh
Query executed successfully.
LENOVO\SQLEXPRESS (15.0 RTM) LENOVO\ELCOT (52) coffee shop 00:00:00 10 rows


SQLQuery1.sql - LENOVO\SQLEXPRESS.coffee shop (LENOVO\ELCOT (52))* - Microsoft SQL Server Management Studio
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Object Explorer
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dbo.sales_person
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Security
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PolyBase
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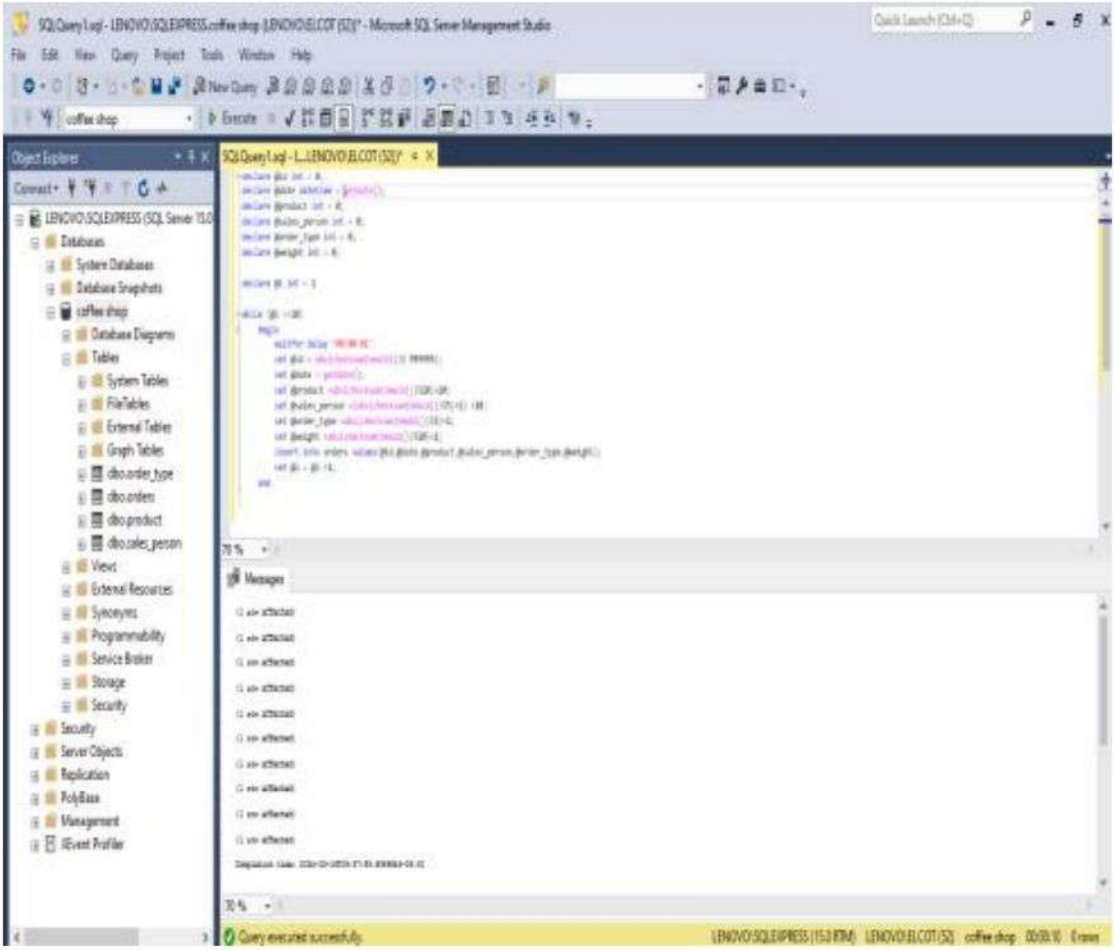
XEvent Profiler
SQLQuery1.sql - L...LENOVO\ELCOT (52))*
insert into product values('1','Coffee',3);
insert into product values('2','Tea',3.1);
insert into product values('3','Drinking Chocolate',4.5);
insert into product values('4','Bread',2);
insert into product values('5','Donuts',3.1);
insert into product values('6','Cookies',3);
insert into product values('7','Biscuits',2);
insert into product values('8','Cake',2);
insert into product values('9','Pastries',4.25);
insert into product values('10','Drinking Chocolate',3.5);
insert into product values('11','Pies',2.55);
insert into product values('12','Rolls',3.5);
insert into product values('13','Muffins',3);
insert into product values('14','Coffee',3.1);
insert into product values('15','Tea',3);
insert into product values('16','Drinking Chocolate',3);
insert into product values('17','Bread',3.75);
insert into product values('18','Donuts',2.5);
Messages
(1 row affected)
(1 row affected)
(1 row affected)
(1 row affected)
(1 row affected)
(1 row affected)
(1 row affected)
Query executed successfully.
LENOVO\SQLEXPRESS (15.0 RTM) LENOVO\ELCOT (52) coffee shop 00:00:00 0 rows
Ready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XEvent Profiler
SQLQuery1.sql - L...LENOVO\ELCOT (52))*
Messages
Query executed successfully.
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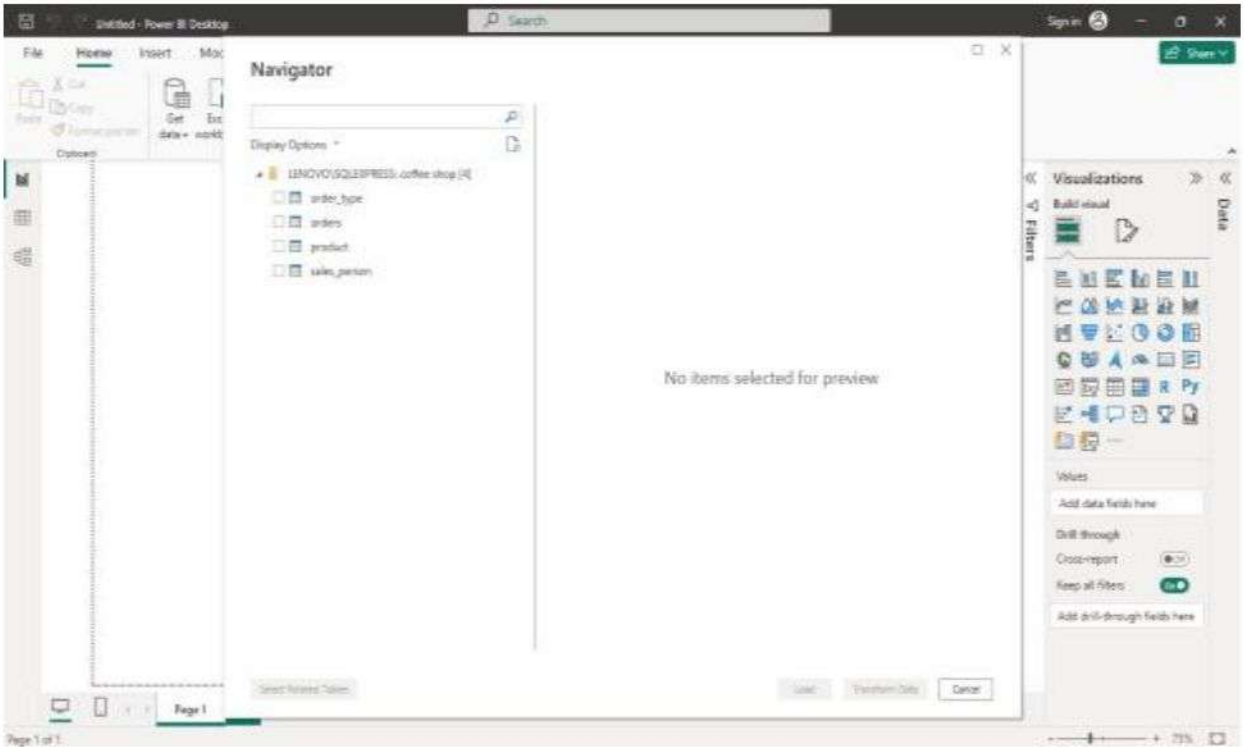
Untitled - Power BI Desktop
Navigator
Display Options
order_type
orders
product
sales_person
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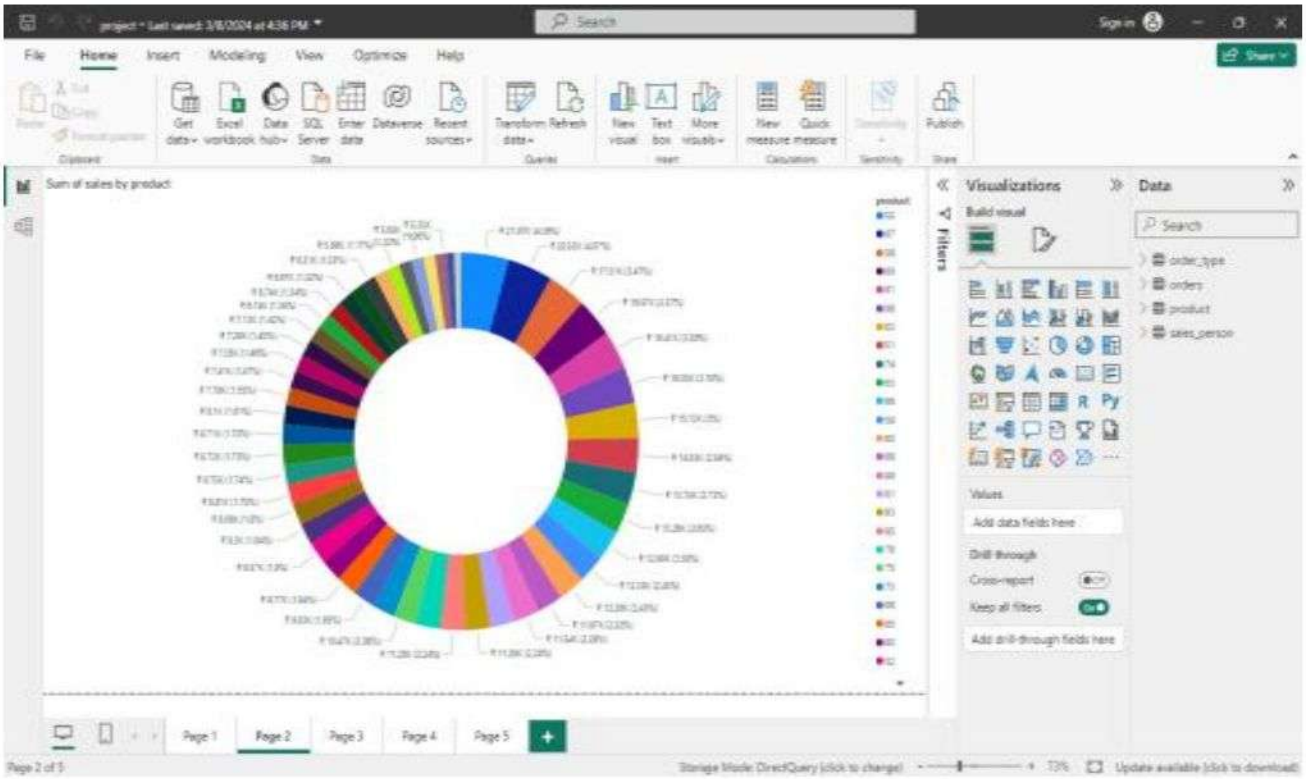
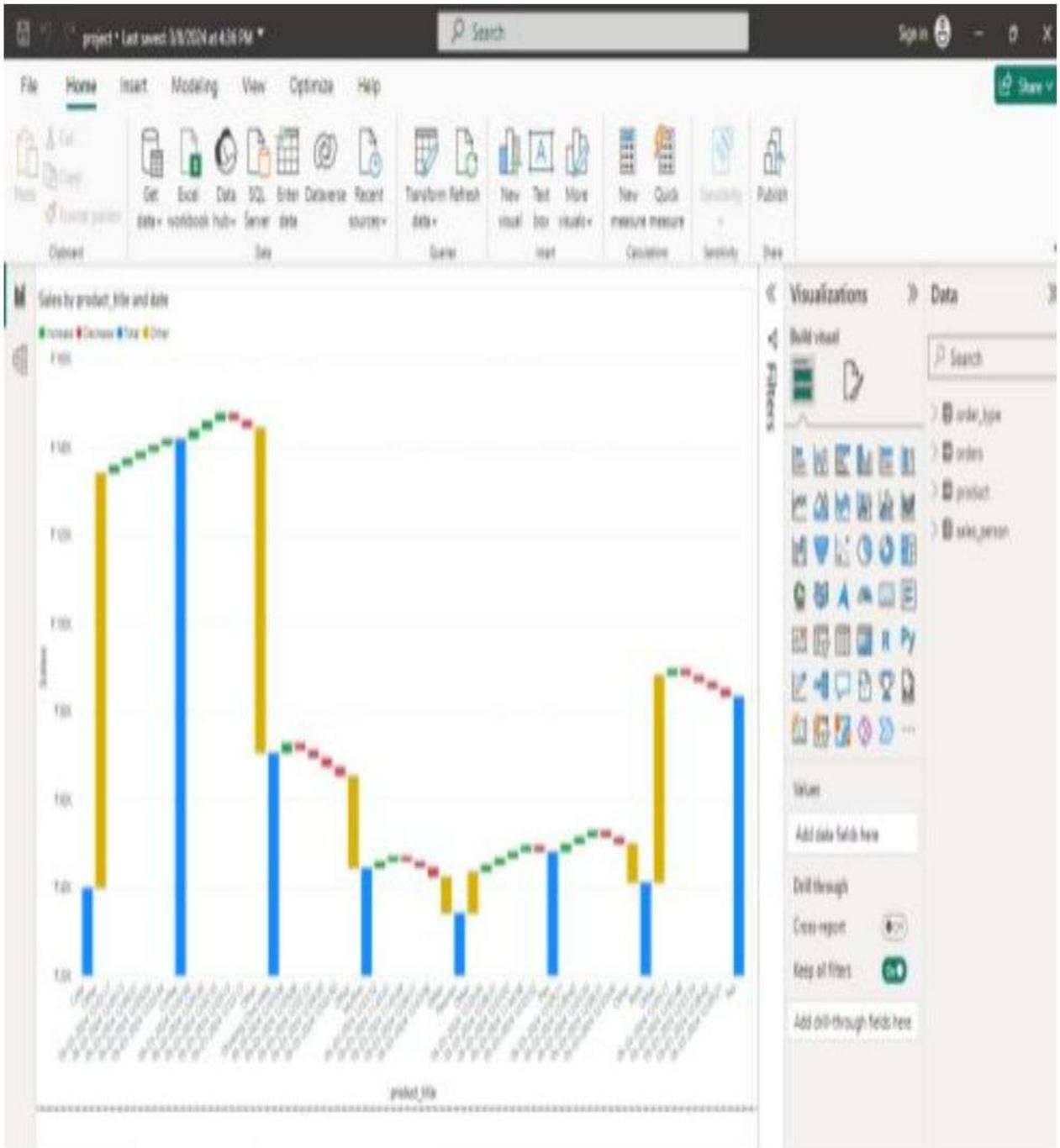
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Page 1


File
Home
Insert
Modeling
View
Optimize
Help
Sum of sales by product
Visualizations
Build visual
Data
order_type
orders
product
sales_person
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Page 1
Page 2
Page 3
Page 4
Page 5


File
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Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
order_type
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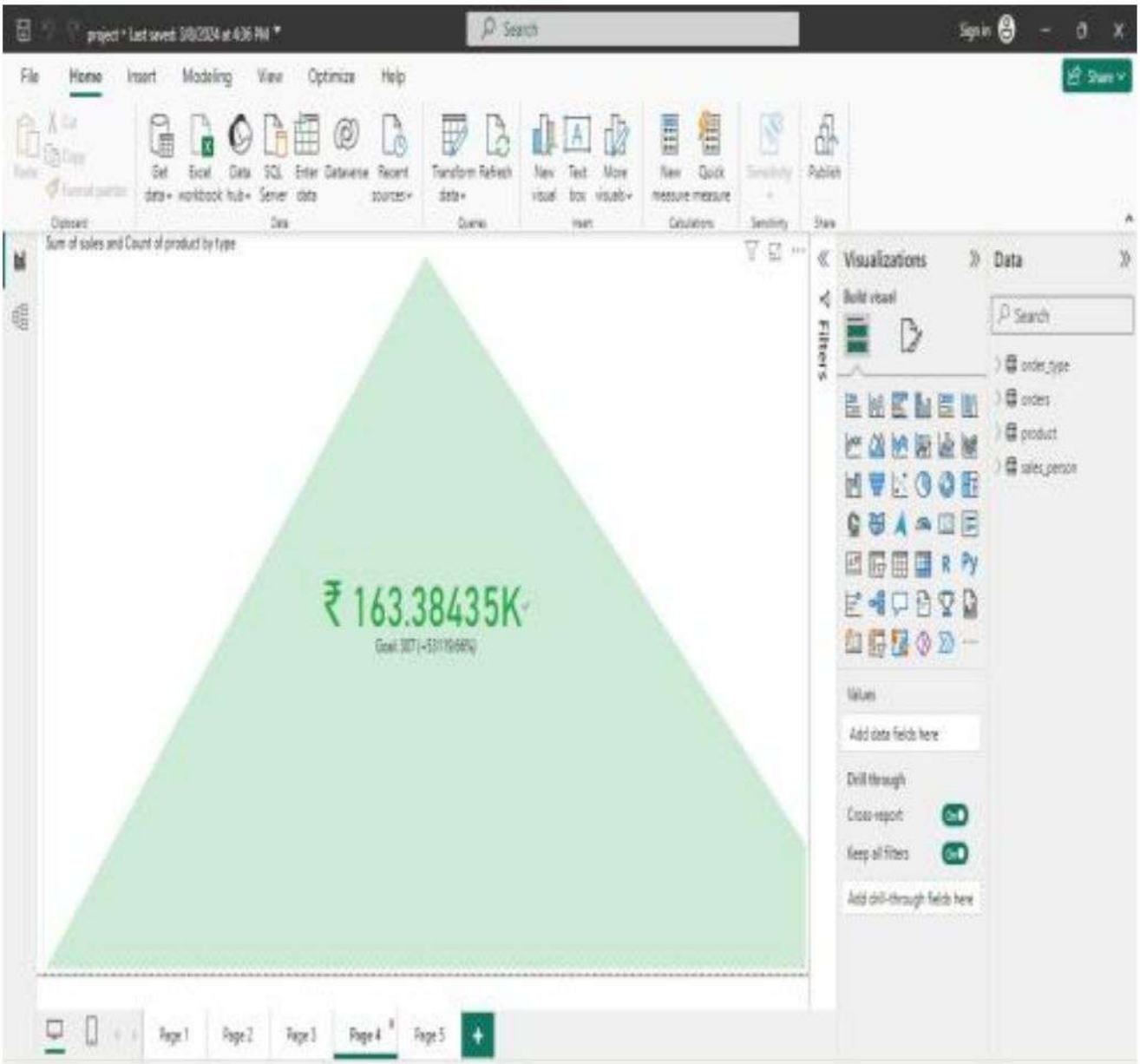
product
sales_person


File
Home
Insert
Modeling
View
Optimize
Help
Sum of sales and Count of product by type
₹ 163.38435K
Visualizations
Data
Build visual
order_type
orders
product
sales_person
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Page 1
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Page 3
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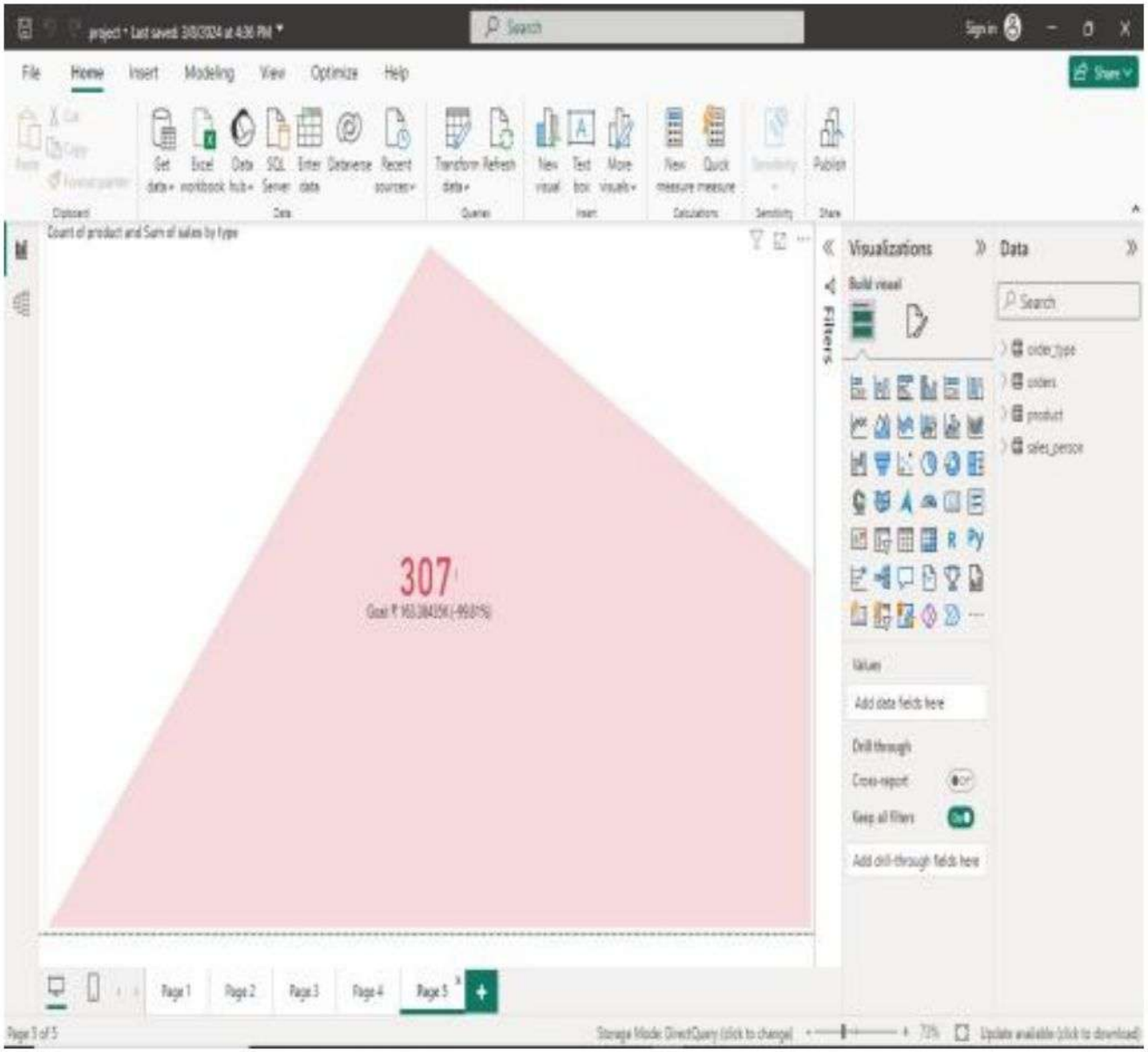
Page 5


File
Home
Insert
Modeling
View
Optimize
Help
Count of product and Sum of sales by type
307
Visualizations
Data
Filters
Build visual
order_type
orders
product
sales_person
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Page 1
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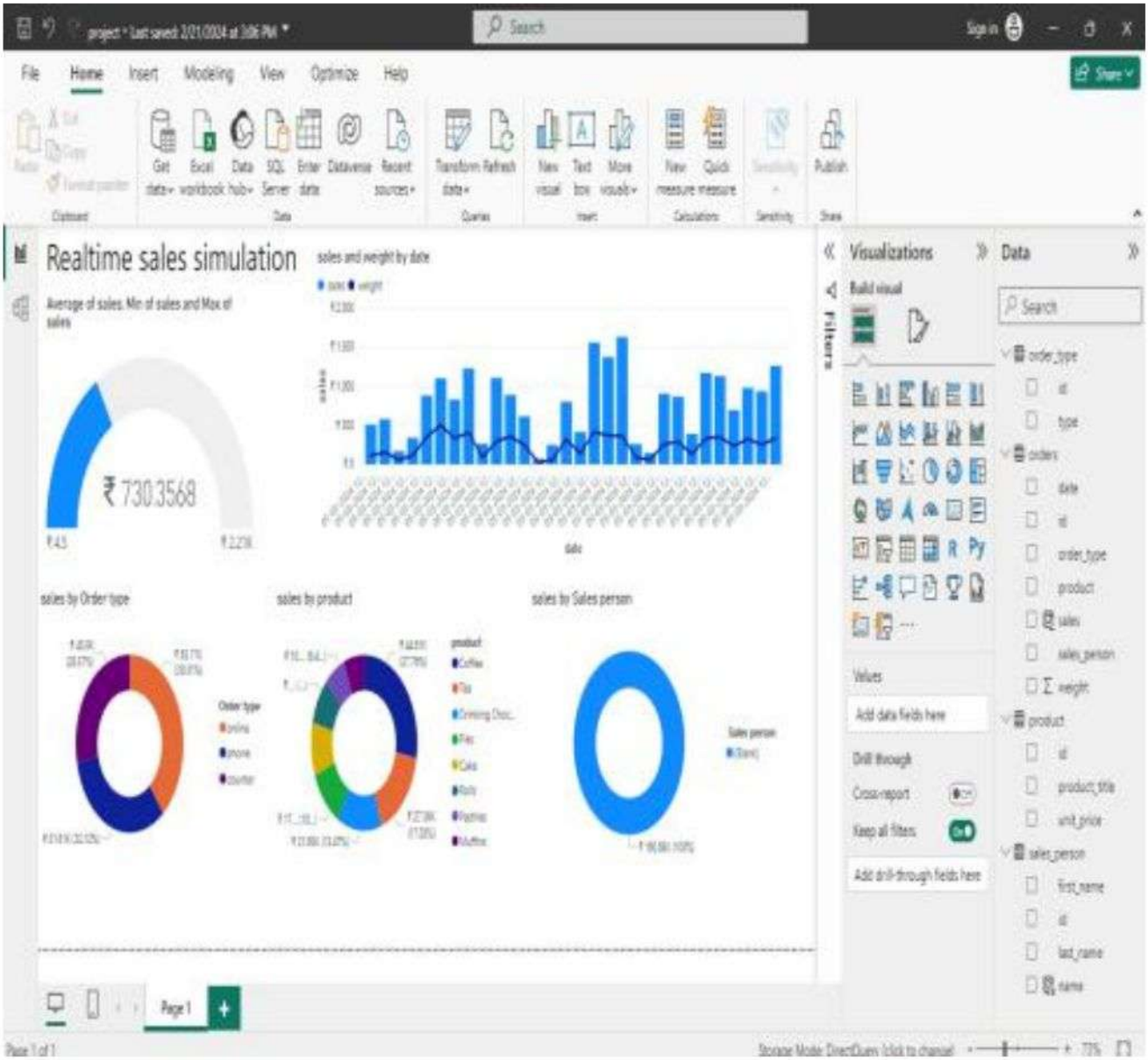
Realtime sales simulation
Average of sales, Min of sales and Max of sales
₹ 730.3568
sales and weight by date
date
sales by Order type
sales by product
sales by Sales person
Visualizations
Data
Filters
Page 1